- Given that the spin has been prepared in the state $|A\rangle$, and that the apparatus is oriented along z, the quantity $\alpha_u^* \alpha_u$ is the probability that the spin would be measured as $\sigma_z = +1$. In other words, it is the probability of the spin being *up* if measured along the z axis.

- Likewise, $\alpha_d^* \alpha_d$ is the probability that σ_z would be *down* if measured.

The α values, or equivalently $\langle u|A\rangle$ and $\langle d|A\rangle$, are called probability amplitudes. They are themselves not probabilities. To compute a probability, their magnitudes must be squared. In other words, the probabilities for measurements of *up* and *down* are given by

$$P_u = \langle A|u\rangle\langle u|A\rangle$$

$$P_d = \langle A|d\rangle\langle d|A\rangle. \qquad (2.2)$$

Notice that I have said nothing about what σ_z is before it is measured. Before the measurement, all we have is the vector $|A\rangle$, which represents the potential possibilities but not the actual values of our measurements.

Two other points are important: First, note that $|u\rangle$ and $|d\rangle$ are mutually orthogonal. In other words,

$$\langle u|d\rangle = 0$$

$$\langle d|u\rangle = 0. \qquad (2.3)$$

The physical meaning of this is that, if the spin is prepared *up*, then the probability to detect it *down* is zero, and vice versa. This point is so important, I'll say it again: Two orthogonal states are physically distinct and